

TASK 9: Backing up and recovery in databases

Perform following backup and recovery scenarios.

- Recovering a NOARCHIVELOG Database with Incremental Backups
- Restoring the server Parameter File
- Performing Recovery with a Backup control File

Scenario 1: Recovering a NOARCHIVELOG Database with Incremental Backups

-- step 1: Backup Database

```
BACKUP DATABASE [database_name] TO DISK = 'backup_file.bak' WITH  
NOFORMAT, NOINIT, NAME = 'Full Database Backup', SKIP, REWIND,  
NOUNLOAD, STATS = 10
```

-- step 2: Create Incremental Backup

```
BACKUP DATABASE [database_name] TO DISK = 'incremental_backup.bak'  
WITH DIFFERENTIAL, NOFORMAT, NOINIT, NAME = 'Incremental Database  
Backup', SKIP, REWIND, NOUNLOAD, STATS = 10
```

-- step 3: simulate Data Loss

-- Intentionally delete or modify data

-- step 4: Restore Database

```
RESTORE DATABASE [database_name] FROM DISK = 'backup_file.bak'  
REPLACE
```

-- step 5: Apply Incremental Backup

```
RESTORE DATABASE [database_name] FROM DISK = 'incremental_backup.bak'  
WITH REPLACE
```

-- step 6: Recover Database

```
RECOVER DATABASE [database_name]
```


Scenario-2: Restoring the server Parameter File (SPFILE)

--step 1: Backup SPFILE

BACKUP SERVER PARAMETER FILE TO FILE = 'spfile.bak';

--step 2: simulate SPFILE loss

--Delete or modify SPFILE.

--step 3: Restore SPFILE

STARTUP MOUNT

RESTORE SERVER PARAMETER FILE FROM FILE = 'spfile.bak';

SHUTDOWN

STARTUP

Scenario 3: Performing Recovery with a Backup control File

--step 1: Backup Control File

BACKUP CONTROLFILE TO FILE = 'controlfile.bak';

--step 2: simulate Control File loss

--Delete or modify control file.

--step 3: Restore Control File

STARTUP MOUNT

RESTORE CONTROLFILE FROM FILE = 'controlfile.bak';

ALTER CONTROLFILE REUSE;

--step 4: Recover Database

RECOVER DATABASE USING BACKUP CONTROLFILE;

Step 5: Open Database

ALTER DATABASE OPEN RESETLOGS;

SQL Server Commands:

- BACKUP DATABASE

- RESTORE DATABASE

- RECOVER DATABASE

- ALTER DATABASE
- BACKUP SERVER PARAMETER FILE
- RESTORE SERVER PARAMETER FILE
- BACKUP CONTROLFILE
- RESTORE CONTROLFILE

VEL TECH	
EX NO.	9
PERFORMANCE (5)	✓
RESULT AND ANALYSIS (5)	✓
VIVA VOCE (5)	✓
RECORD (5)	✓
TOTAL (20)	20
SIGN WITH DATE	18/09/20

RESULT:- Thus, the backing up and recovery in databases was successfully executed.